

Self-Attention conditional generative adversarial network optimised with crayfish optimization algorithm for improving cyber security in cloud computing

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ABSTRACT

The decentralized and distributed architecture of cloud computing promotes adoption and growth in various societal domains, including education, government, information technology, business, entertainment. Cloud computing (CC) makes a broad range of information technologies available. Security and privacy are key challenges in storing big data in the cloud. To overcome this challenge, a Self-Attention Conditional Generative Adversarial Network Optimised with Crayfish Optimization Algorithm for Improving Cyber Security in Cloud Computing (CybS-CC-SACGAN—COA) is proposed in this paper to enhance the safety of CC environment. The input data is amassed from NSL-KDD database. After that, the data is fed to pre-processing segment. The segment of pre-processing eliminates Cloud data termination and missing value replacement using Reformed Phase Conserving Vibrant Range Compression filter. The results of pre-processing serves for feature selection. The optimal features are selected by means of Manta Ray Foraging Optimization Algorithm (MRFOA). Cloud data is categorized as normal and abnormal data, like Denial-of-Service (DoS), Probe, Remote to local attack (R2L), User to root attack (U2R) by the help of SACGAN. Crayfish Optimization Algorithm (COA) is proposed to optimize the SACGAN classifier that classifies anomaly data precisely. The proposed CSCC-SACGAN—COA technique is activated in Python under some metrics. The proposed CSCC-SACGAN—COA approach has attained higher detection accuracy, lower computation time, higher AUC, higher scalability and lower detection error rate compared to the existing methods.

1. Introduction

The term "cloud computing" symbolizes internet-based computing that uses virtual shared servers and workstations to supply platforms, software, infrastructure, and resources. This type of computing is constantly growing (Mayuranathan et al., 2021). Cloud computing is susceptible to attacks, because of its open structure. The success of cloud computing depends on privacy and security (Rajan and Naganathan, 2022; Lou et al., 2021). Traditional security protocols cannot address this issue. Attacks like the Sybil attack and the Denial of Service (DoS) are inevitable (SobinSoniya and Maria Celestin Vigila, 2022). The detection of hostile network activity is among the most important and longstanding cyber security issues. The complexity of security-related issues is rising due to the spread of networks and various technologies

being used (Kholidy, 2021; SobinSoniya and Maria Celestin Vigila, 2022; Srilatha and Shyam, 2021). Furthermore, odd network behavior is misconstrued as a symptom of a compromised device or network. From disseminated computing Cloud computing is progressed the practice of the cloud increases and the number of clients shares single to the cloud resources by using consistent inaccessibility technologies (Prabhakaran and Kulandasamy, 2021; Mehanović et al., 2021; Guezzaz et al., al.). It causes an improvement of traffic network malevolent occurrences becoming more extensive and difficult (Vu et al., 2022). Cloud computing contains many benefits such as; scalability of instance, quick elasticity, dignified services the most important key feature are the assistances of cost savings to enterprises and also using on-demand services in virtual manner through internet (Selvapandian and Santhosh, 2021; Krishnaveni et al., 2021). It is difficult to predict the assets in

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cloud and also the Communication between them, hence quantity of intrusions increasing gradually (Rajagopal et al., 2021). Nowadays, reports of statistical shows excellent number of securities violates occur in cloud computing of virtual network layer (Mondal and Goswami, 2021; Ghosh et al., 2021). Intrusion detection scheme (IDS) is the widely used part of computer system safety processes for safeguarding cloud environments from cyber-attacks (El Kafhali et al., 2022; Tissir et al., 2021; Hassaoui et al., 2024; Hassaoui et al., 2023). To keep track of the development of computer-related crimes, intrusion detection systems (IDS) employ a variety of reaction mechanisms to find weaknesses, spot illicit activity, and take preventative action (Turcato et al., 2021). Cyber security systems (CSS) in platform of cloud tools and techniques are available to identifying attacks on cloud setup (Qureshi et al., 2021; Ogwara et al., 2021; Samunnisa et al., 2023). Due to heavy traffic and its changes of behavior traditional CSS is turn out ineffective and inefficient. To form a good security solution CSS use firewall (Dehghani et al., 2022).

Modern and sophisticated technologies are essential to estimate the massive volume of data created from the interactions among the devices of cloud computing systems because of the complexity and size of the distributed infrastructure of cloud computing networks. Real-time analysis of the complex data generated is necessary for the development of robust intrusion detection systems. Therefore, deep learning techniques are promising for the analysis of large data, consisting of traffic flows in cloud networks for indicators of intrusions, even if they have not been detected before. These are motivated to do this research.

An anomaly-based network intrusion detection scheme is proposed that can track and examine the network traffics flow targeting cloud environments. The nature of these traffics should be reported to the network administrator so that any obtrusive network connections can be dropped and blocked. SACGAN is considered as a classifier of the network connections. The most pertinent network features are chosen using the binary-based MRFOA, and the SACGAN control parameters are adjusted using the standard-based COA. The proposed system is assessed using network data from the benchmark NSL-KDD dataset. The outcomes shows that the proposed method is identified the intrusive network connections with maximal accuracy with less false alarm rates (FARs).

The primary contributions of this paper are abridged below:

- CyBS-CC-SACGAN—COA is proposed for protecting CC environment.
- The feature selection and categorization method for IDS is explored. Proposed a feature selection strategy for intrusion identification based on the MRFOA.
- The proposed method categorizes network attacks by employing SACGAN, which classifies the data into DoS, Probe, R2L, U2R, Normal (no attack).
- Crayfish Optimization Algorithm (COA) is proposed to optimize the SACGAN classifier, which precisely categorized by the CC Intrusion.
- The efficacy of the proposed method is analyzed with existing CyBS-CC-BPNN-ArBCOA, CyBS-CC-DKNN, CyBS-CC-DNN-GTOA and CyBS-CC-WDBiLSTM-GWO methods.

Remaining paper is structured as below: the recent studies are depicted in section II, section III describes the proposed method, section IV proves the result, section V presents the conclusion.

2. Related work

The recent works related to deep learning based intrusion identification of cloud environment are detailed in this section.

Wen et al., (Wen, 2022) have suggested CC intrusion detection technology basis on BP-NN. Initially, intrusion cloud dataset was taken from KDD cup dataset and it was given to Cyber security system. In Cyber security system, back propagation neural network were exploited for categorizing the cloud data as normal or anomalous data.

Consequently, the back propagation neural network was enhanced by utilizing the artificial bee colony optimization technique. It presents better results under F-Score, but it increases the computation time.

Mayuranathan et al., (Mayuranathan et al., 2022) have presented efficient optimal security system for intrusion recognition in CC utilizing hybrid deep learning model. Mainly the cloud data was collected and transferred to the pre-processing section. The Pre-processing segment, improved heap optimization technique were exploited to improving the data quality by eliminating unwanted data. The features of optimal were fed Kronecker of DNN classify the intruders. It presents better results under sensitivity, but it decreases the precision value.

Balamurugan et al., (Balamurugan et al., 2022) have presented Network optimization utilizing defender scheme in cloud computing safety basis intrusion identification scheme along game theory DNN (IDSGT-DNN). Primarily, the intrusion cloud dataset was taken and given to Cyber security system. In Cyber security system, DNN was exploited for the categorization of cloud data as normal and anomalous data. In general, DNN has not rapid any adoption of optimization techniques to measure appropriate parameters to ensure a certain categorization to computing of cloud intrusion. So, game theory optimization technique was exploited to enhance the DNN. The presented method provides better results in terms of accuracy, but it provides low area under curve value.

Geetha and Deepa, (Geetha and Deepa, 2022) have presented FKPCA-GWO WDBiLSTM classifier IDS in environments of cloud. Initially, the intrusion cloud dataset was collected and sent to the Fisher kernel based Principal of Component Analysis dimensionality reducing algorithm by choosing the optimal features. The chosen optimal features were provided to gray wolf optimizer established weight dropped BiLSTM classifier to classifying the cloud intruders. It provides better result in terms of specificity, but it increases the detection error rate.

Kumar et al., (Kumar et al., 2022) have presented intrusion identification with prevention of CC: From perspective of deep learning. Initially, intrusion cloud dataset was taken and it was given to CSS. In cyber security system, Fuzzy Min Max Neural Networks-Based Scheme was utilized for classification of intruders. It gives better results in term precision but it decreases the sensitivity.

Jain et al., (Jain et al., 2022) have presented Training fuzzy deep neural network along honey badger algorithm (HBA) in CC for intrusion identification. The presented study suggest a novel fuzzy deep neural network (FDNN) for cloud environments called FDNN—HBA was developed in conjunction with the Honey Bader Algorithm to provide privacy-preserving intrusion identification. An efficient training method was used for intrusion identification and categorization using the FDNN model. The presented method provides better sensitivity but increased computation time.

Thilagam and Aruna (Thilagam and Aruna, 2021) have presented Intrusion detection for network based CC by custom RC—NN with optimization. Initially, intrusion of cloud dataset was taken and it given to Cyber security system. RCNN with LSTM and the Ant Lion optimization algorithm were utilized in cyber security systems to classify cloud data as normal and abnormal. The presented method lowers the F-Score but yields better area under the curve results. Table 1 depicts the related studies comparison.

3. Proposed methodology

In this manuscript, the CyBS-CC-SACGAN—COA is proposed to improve the security on cloud computing. The block diagram of CyBS-CC-SACGAN—COA is shown in Fig. 1. It comprises data acquisition, pre-processing, feature selection, cloud cyber security system and optimization. Description of the detailed proposed methodology is given below,

Table 1
Comparison of Related Work.

Author Name	Objective	Methods	Performance Measures	Advantages	Disadvantages
Wen et al., (Wen, 2022)	Cloud Computing intrusion identification depending upon BP-NN.	BP neural network, Artificial Bee Colony Optimization Algorithm (ABCOA).	Accuracy rate, false alarm rate, detection time, error rate, computation time.	It provides high accuracy.	It has high computation time.
Mayuranathan et al., (Mayuranathan et al., 2022)	Proficient optimal safety scheme for intrusion identification in CC under hybrid deep learning mode.	Improved Heap Optimization (IHO), Chaotic Red Deer Optimization (CRDO), Deep Kronecker Neural Network (DKNN)	Accuracy, Precision, latency, false alarm rate, detection time, error rate and computation time.	It provides low latency.	It has high error rate.
Balamurugan et al., (Balamurugan et al., 2022)	Network optimization utilizing defender scheme in cloud computing safety basis intrusion identification scheme along game theory DNN.	Game Theory Deep Neural Network	Accuracy, detection rate, precision, F-Score, AUC, FPR.	It offers higher F-Score.	It has low AUC.
Geetha and Deepa (Geetha and Deepa, 2022)	fkpca-gwo wdbilstm classifier ids in environments of cloud.	gray wolf optimizer, bilstm classifier, and pca dimensionality reduction technique based on fisher kernel	accuracy, precision, sensitivity, specificity, F-measure, MCC, kappa, aROC	It provides high precision.	It has low MCC.
Kumar et al., (Kumar et al., 2022)	From a deep learning, cloud computing intrusion finding and prevention	Fuzzy Min Max Neural Networks, fuzzy min-max learning approach.	Accuracy rate, false alarm rate, detection time, error rate and computation time.	It provides high accuracy.	It has high computation time.
Jain et al., (Jain et al., 2022)	Training FDNN along HBA for intrusion identification in CC.	FDNN, HBA	Accuracy, Precision, latency, false alarm rate, detection time, error rate and computation time.	It provides low latency.	It has high latency.
Thilagam and Aruna [32]	Intrusion finding for network based CC through custom RC—NN with optimization.	Recurrent Convolutional Neural Network, Ant Lion optimization, CNN, LSTM	Error rate, true positive rate, true negative rate, precision and recall.	It provides high recall.	It has low accuracy.

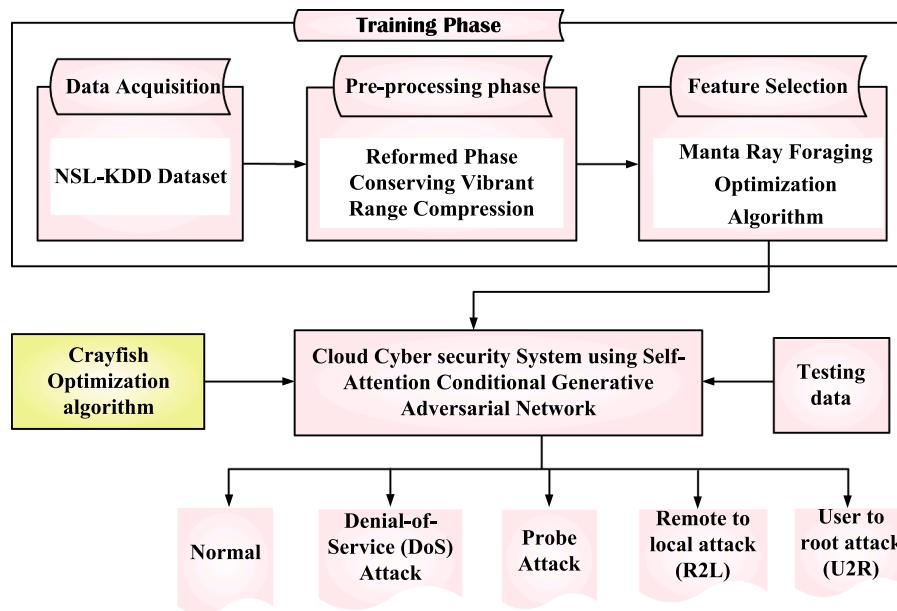


Fig. 1. Proposed CyBS-CC-SACGAN—COA Technique.

3.1. Data acquisition phase

Initially, input cloud data are gathered from NSL-KDD database (NSL-KDD dataset). This database is an improved version of KDD cup 1999 database used in intrusion identification tests. Network Security Laboratory (NSL-KDD) benchmark dataset is subjugated that demands of persuaded conventional cloud data cluster through client device which encompasses copious interference that are playacts in the cloud. There are four main categories of abnormal and several normal records in the NSL-KDD dataset. In this work, interference forms are categorized into DoS, Probe, R2L attack, U2R, Normal. There are 41 features from various data packet and 1 class label for every data packet. Table 2 tabularizes NSL-KDD database.

3.1.1. Denial-of-Service (DoS) attack

DoSs effort blackout the traffic flow to and from the terminus system. IDS are inundated by unexpected range of traffic that the system cannot blackout to guard the concern. DoS exhibitions incorporate Back, Land, Neptune, Apache2, Mailbomb, etc.

3.1.2. Probe

Probe is an exhibition that exertions to contract data from a cloud computing environment. Probe intervention integrates Satan, ipsweep, Nmap, Portsweep, saint and msan.

3.1.3. Remote to local attack (R2L)

R2L is an indicator that endeavours to acquire restricted communication with a remote machine. An invader could not locally investiture to the cloud computing environment and exertions to "hack" its network

Table 2
Features of NSL-KDD database.

Feature Name	Category
Duration	Continue
Type of Protocol	Categoric
Service	Categoric
Flag	Categoric
Src Bytes	Continue
Dst Bytes	Continue
Land	Binary
Failed Fragment	Distinct
Urgent	Distinct
Hot	Continue
Num Fault Logins	Continue
Log In	Binary
Num Compromised	Continue
Root Shell	Binary
Su Attempt	Distinct
Num Root	Continue
Num File Creations	Continue
Num Shells	Continue
Num Accessing Files	Continue
Num Outbound Cmds	Continue
Is Hot Logins	Binary
Is Guest Login	Binary
Count	Distinct
Srv Count	Distinct
Error Rated	Distinct
SrvError Rated	Distinct
Error Rated	Distinct
SrvError Rated	Distinct
Same Srv Rate	Distinct
Diff Srv Rated	Distinct
Rate of Srv Diff Host	Distinct
Count of Dst Host	Distinct
Count of Dst Host Srv	Distinct
Dst Host Similar Srv Rated	Distinct
Dst Host Dissimilar Srv Rated	Distinct
Dst Host Similar Src PortRated	Distinct
Dst Host Srv Dissimilar HostRated	Distinct
Dst Host SerrorRated	Distinct
Dst Host SrvSerrorRated	Distinct
Dst Host RerrorRated	Distinct
Dst Host SrvRerror Rated	Distinct
Class	Categorical
Difficulty Level	Distinct

investiture. R2L realizes Guess_passwd, Ftp_write, Imap, http tunnel, named, Phf, send mail, snmpgetattack, snmpguess, xlock, xsnoop, Multi hop, Ware master, Spy and Ware client.

3.1.4. User to root attack (U2R)

U2R is an incidence unambiguously provoked by distinct user ac-

Vibrant Range Compression (RPCVRC) (Osman and Yap, 2020) is designated. Here, RPCVRC filter were exploited for exterminating the Cloud data redundancy and missing value restoration. The output of the pre-processing Cloud data is determined using the Eq. (1)

$$\text{Pre-processed Cloud Data} = \log(\text{Cloud Data} + 1) \cdot \sin(\phi) \quad (1)$$

By the Eq. (1), RPCVRC filter exterminates the data redundancy including restoration of the missing value of NSL-KDD dataset. The pre-processing cloud data is provided to the feature selection.

3.3. Feature selection under manta ray foraging optimization algorithm

This section describes the feature selection method. The output of pre-processing is provided to the feature selection as input. This program is inspired by the manta ray cognitive behaviors. This work develops the three forms of foraging behaviors: somersault, cyclone and chainforaging provide an efficient optimization paradigm for feature selection.

3.3.1. Stepwise procedure of manta ray foraging optimization algorithm

Manta rays are fancy animals despite their terrific appearance. They are among the biggest marine animals. Manta rays are flat-bottomed animals with two pectoral fins that permit to swim gracefully like birds do when they fly. MRFOA is inspired by three forms of foraging behaviors: somersault, cyclone, and chain. The stepwise procedure of the MRFOA is deliberated below.

3.3.1.1. Step 1: initialization. Initially, manta ray population are initialized for attaining optimal features from NSL-KDD dataset and expressed using Eq. (2)

$$i_t(g) = i_{lwr} + rd * (i_{upr} - i_{lwr}); \text{ for } t = 1, 2, \dots, \text{Population}_w \text{ and } g = 1 \quad (2)$$

3.3.1.2. Step 2: random generation. After initializing, the input-parameters created randomly. The ideal fitness values are deemed with regard to its clear hyper parameter situation.

3.3.1.3. Step 3: fitness function estimation. This is articulated in Eq. (3),

$$\text{Fitness function} = [\text{Selecting Optimal features}] \quad (3)$$

3.3.1.4. Step 4: chain foraging. In MRFOA, manta rays are able to detect plankton's location and move in its direction. A site's quality rises in proportion to its plankton concentration. In this Chain foraging segment, the plankton positions are remarked by Manta rays and whirl to it. And it is expressed based on the succeeding Eq. (4)

$$i_t^{\text{dim}}(g+1) = \begin{cases} i_t^{\text{dim}}(g) + rd \cdot (i_{best}^{\text{dim}}(g) - i_t^{\text{dim}}(g)) + \partial \cdot (i_{best}^{\text{dim}}(g) - i_t^{\text{dim}}(g)); & t = 1 \\ i_t^{\text{dim}}(g) + rd \cdot (i_{t-1}^{\text{dim}}(g) - i_t^{\text{dim}}(g)) + \partial \cdot (i_{best}^{\text{dim}}(g) - i_t^{\text{dim}}(g)); & t = 2 \dots \text{Population}_w \end{cases} \quad (4)$$

count, and then exertions to interface the cloud computing environment, as a superuser (root). The invader happenstances to deed exposures in a scheme to advance root access/rights. U2R encompasses buffer overflow, loader module, rootkit, perl, ps, sqlattack, and xterm.

At that time, the input cloud data are specified as the input of pre-processing phase.

3.2. Pre-processing using reformed stage conserving vibrant range compression

In this section, Pre-processing utilizing Reformed Stage Conserving

where $i_t^{\text{dim}}(g)$ characterizes of place t^{th} individual at the time gin the dimensiondim, rd elucidates the random value lies amongst 0 and 1, $i_{best}^{\text{dim}}(g)$ exposes the high attentiveness plankton.

3.3.1.5. Step 5: cyclone foraging. Once manta ray determines a plankton predicament in deep water make protracted foraging chain and warped transferral to the food. This Twister searching segment is enunciated based on the Eq. (5)

$$i_t^{\dim}(g+1) = \begin{cases} i_{best}^{\dim} + rd \cdot (i_{best}^{\dim}(g) - (i_t^{\dim}(g))) + \eta \cdot (i_{best}^{\dim}(g) - i_t^{\dim}(g)); & t = 1 \\ i_{best}^{\dim} + rd \cdot (i_{t-1}^{\dim}(g) - i_t^{\dim}(g)) + \eta \cdot (i_{best}^{\dim}(g) - i_t^{\dim}(g)); & t = 2 \dots Population_w \end{cases} \quad (5)$$

where *Iteration* epitomizes the maximum number of iterations; *rd* represents random value also values are lies between 0 and 1. Then the food location is perceived as a pivot.

3.3.1.6. Step 6: termination. The feature selection is based on MRFO Algorithm iteratively repeat step 3 until fulfilling $us = us + 1$ halting criteria. If all the process is achieved it will select the optimal features. The selected features by MRFOA are illustrated in table 3.

3.4. Cloud computing cyber security systems using self-attention conditional generative adversarial network

Cloud Computing Cyber Security System utilizing SACGAN (Gomathi et al., 2023; Zhao et al., 2020; Li et al., 2022) is discussed in this section. The chosen features are fed input to categorization stage. The data is classified into DoS, Probe, R2L, U2R and Normal. SACGAN allows attention-driven, long-range dependency modeling for data classification tasks and Cloud Computing Cyber Security Systems. Primarily, loss of function conditional Generative Adversarial Network are articulated with the help of succeeding Eq. (6)

$$Loss_{CdGAN}(Gr, Ds) = E_{(a,b)}[\log Ds(a, b)] + E_{(a)}[\log (1 - Ds(a, Gr(a)))] \quad (6)$$

where *a* depicts the nominated features of cloud data; *b* depicts the final classification of cloud data; *Gr* and *Ds* depicts the generator and discriminator from SACGAN classifier. The objective function of SACGAN classifier are given in the following Eq. (7)

$$\argmin_{Gr}(\argmax_{Ds} Loss_{CdGAN}(Gr, Ds) + \Re * Loss_1(Gr)) \quad (7)$$

where $Loss_1(Gr) = E_{(a,b)}(\|b - Gr(a)\|_1)$ embodies the $Loss_1$ among the ground truth and generated cloud data, $\Re$ epitomizes the regularization tenure for the $Loss_1$. Then self-attention is applied in the conditional Generative Adversarial Network for improving the accuracy of Cloud

computing Cyber security systems. The cloud features $mapCon \in \mathbb{R}^{No \text{ of Channel} \times Total \text{ cloud Data}}$ from one transitional hidden layer of SACGAN classifier was transformed into two cloud features spaces, for attaining self-attention $x(a) = Weight_x * a$ and $y(a) = Weight_y * a$. Correspondingly, $f_{p,q}$ can be deliberate based on the succeeding Eq. (8)

$$f_{p,q} = x(a_q)^T y(a_p) \quad (8)$$

Attention layer output is assessed in expressed Eq. (9),

$$O_q = K \left[\sum_{p=1}^{Total \text{ cloud Data}} \beta_{p,q} * z(a_p) \right] \quad (9)$$

where *K* is accessed based on the succeeding Eq. (10),

$$K(a) = Weight_K * a \quad (10)$$

The Self Attention output is appraised depending on given eqn (11),

$$b_q = \chi * O_q + a_q \quad (11)$$

where χ illustrates the scaling constraint. Then to alleviate the SACGAN classifier training, Feature alike loss is exploited only in the generator. The feature alike loss can be appraised based on Eq. (12),

$$Loss_{Fr \text{ alike}}(Gr, Ds) = E_{(a,b)} \sum_{p=1}^{Total} \frac{1}{Total \text{ Cloud Data}_p} \|Ds^p(a, b) - Ds^p(a, Gr(a))\|_1 \quad (12)$$

where Ds^p is the p^{th} feature map layer; *Total* depicts the total layers discriminator.

Then the objective function of SACGAN classifier with error loss of the cloud data are given in the following Eq. (13)

$$\argmin_{Gr}(\argmax_{Ds} Loss_{CdGAN}(Gr, Ds) + \Re * Loss_1(Gr) + \alpha * Loss_{Fr \text{ alike}}(Gr, Ds) + \eta * Loss_{error}(Gr)) \quad (13)$$

where η depicts the regularization tenure constraint. The objective function of SACGAN classifier with $Loss_2$ norm regularizations to generating and discriminator are given in the following Eq. (14)

$$\argmin_{Gr}(\argmax_{Ds} (Loss_{CdGAN}(Gr, Ds) - K_{Ds} * Loss_2(Ds)) + \Re * Loss_1(Gr) + \alpha * Loss_{Fr \text{ alike}}(Gr, Ds) + \eta * Loss_{error}(Gr) + K_{Gr} * Loss_2(Gr)) \quad (14)$$

where K_m and K_n control the importance of $Loss_2$ norm regularizations of generator and discriminator. By this, proposed SACGAN classifier accurately categorizes the invaders as DoS, Probe, R2L, U2R and Normal. The SACGAN classifier uses artificial intelligence based optimization technique based on artificial intelligence is employed due to its suitability, accessibility etc. COA is utilized to improve SACGAN classifier and find the ideal parameters. Generally, few techniques are employed to limit the formation such as manual random, grid investigations. These explorations disperse its unique weakness depend on repetition time; it lacks an in-depth examination of deceitfully constructed. Thus, the COA is used to overcome the issues.

3.5. Optimization using crayfish optimization algorithm

Here, $\Re$, α and η parameter of SACGAN is optimized by COA. COA mimics the competitiveness, foraging behaviors of crayfish. COA overcomes the manual random and grid explorations problems. To strike a

Table 3
Nominated Features using Manta Ray Foraging Optimization Algorithm.

Feature Name	Type
Protocol Category	Category
Service	Category
Land	Binary
Incorrect Fragment	Discrete
Hot	Continual
Logged In	Binary
Num Root	Continual
Count	Discrete
SerrorRated	Discrete
SrvErrorRated	Discrete
RerrorRated	Discrete
SrvRerrorRated	Discrete
Similar SrvRated	Discrete
Distinct Srv Rate	Discrete
Amount of Dst Host	Discrete
Count of Dst Host Srv	Discrete
Dst Host Same SrvRated	Discrete
Dst Host Diff Srv Rated	Discrete
Dst Host Srv Diff Host Rated	Discrete
Dst Host Serror Rated	Discrete
Dst Host SrvSerror Rated	Discrete
Dst Host Rerror Rated	Discrete
Dst Host SrvRerror Rated	Discrete
Class	Categorical

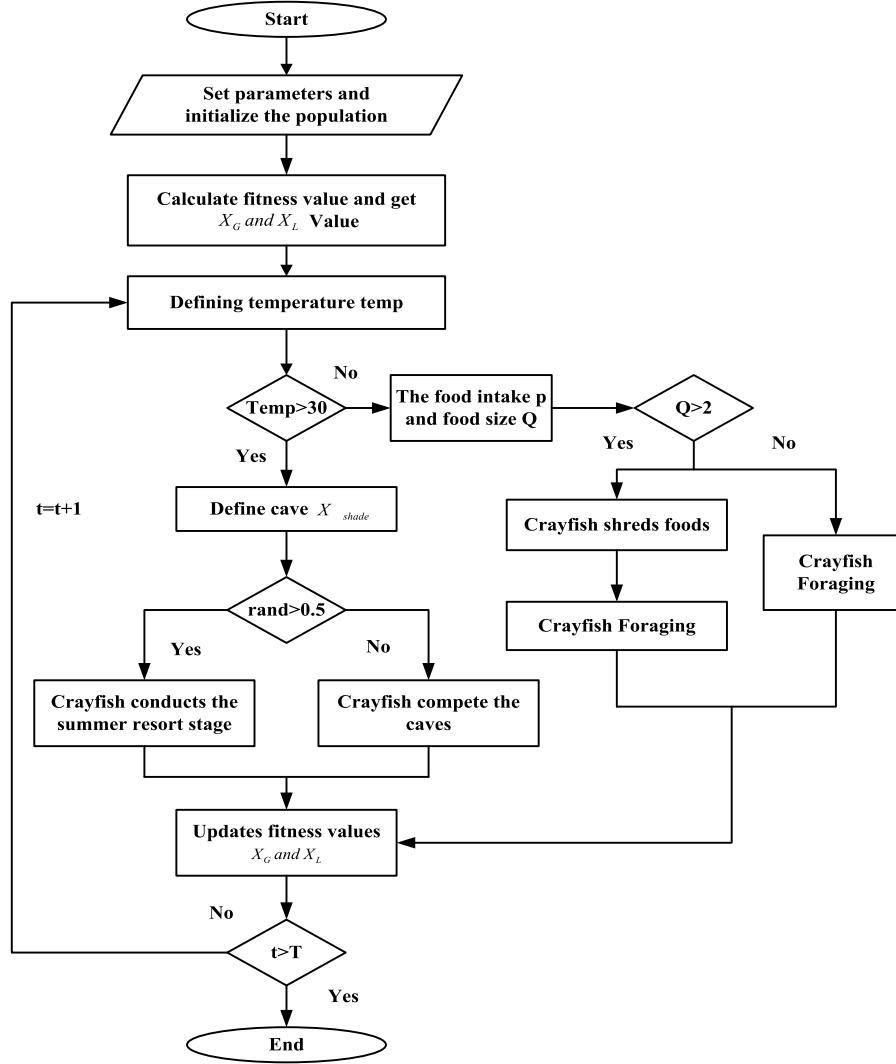


Fig. 2. Flowchart of COA for optimizing SACGAN classifier parameters.

balance between algorithmic exploration and exploitation, the three behaviors are split into three distinct stages. The summer resort, the competition, and forage stage are the three phases. The COA's exploratory stage is represented by the summer resort stage. The COA's exploitation stage is represented by the stages of competition and foraging.

3.5.1. Stepwise processing of COA for developing the SACGAN classifier

The foraging, summertime vacation and competitive behavior of crayfish serve as inspiration for COA. The exploitation stages of COA are the foraging and competitive stages, while the exploration stages are the summer resort stages. The flowchart of crayfish optimization algorithm is given in Fig. 2.

3.5.1.1. Step 1: initialization. Every crayfish is $1 \times \text{dim}$ matrix on multiple dimensional optimization issue. The problem solution is represented by each column matrix. The initialization of COA algorithm is given in Eq. (15)

$$X = \begin{bmatrix} X_{1,1} & \cdots & X_{1,j} & \cdots & X_{1,\text{dim}} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ X_{i,1} & \cdots & X_{i,j} & \cdots & X_{i,\text{dim}} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ X_{N,1} & \cdots & X_{N,j} & \cdots & X_{N,\text{dim}} \end{bmatrix} \quad (15)$$

Let X denotes positioning of initial populace, N denotes populace

count, dim signifies population dimension, X_{ij} denotes positioning of individual i at j dimension, X_{ij} is represented in Eq. (16),

$$X_{ij} = lb_j + (ub_j - lb_j) \times \text{rand} \quad (16)$$

where lb_j signifies lesser bound of j^{th} dimension, ub_j denotes upper bound of j^{th} dimension, rand implies random number.

3.5.1.2. Step 2: random generation. After initializing, the input-parameters created randomly. The ideal fitness values are considered depend upon its clear hyper parameter condition.

3.5.1.3. Step 3: fitness function estimation. After initialization, this solution is evaluated as well as the objective of the function represents parameter values of $\mathfrak{R}$, α and η regularization tenure constraint from SACGAN classifier. Thus expressed in Eq. (17),

$$\text{Fitness Function} = \text{Optimizing } [\mathfrak{R}, \alpha, \eta] \quad (17)$$

3.5.1.4. Step 4: define temperature and intake of crayfish. The crayfish behavior is impacted by temperature changes and undergoes several phases. Temperature is calculated by given Eq. (18),

$$\text{temp} = \text{rand} \times 15 + 20 \quad (18)$$

where $rand$ indicates random value, $temp$ indicates the temperature in the location of crayfish. For their summer vacation, crayfish will select a cool spot when the temperature rises above 30 °C. When the temperature is right, crayfish will start to feed.

3.5.1.5. Step 5: summer resort stage (Exploration). The temperature is excess when $temp > 30$. In this moment, the crayfish decide to spend the summer vacation in cave. Cave X_{shade} is expressed in Eq. (19),

$$X_{shade} = (X_G + X_L) / 2 \quad (19)$$

where, X_G denotes the ideal position acquired by the count of iterations, X_L denotes the existing population best position. Crayfish fighting caverns happens randomly.

3.5.1.6. Step 6: competition stage (Exploitation). When $temp > 30$ and $rand > 0.5$, it indicates that the cave is of interest to further crayfish. They will now engage in combat to take the cave. The crayfish competes for cave in Eq. (20).

$$X_{i,j}^{t+1} = X_{i,j}^t - X_{z,j}^t + X_{shade} \quad (20)$$

here, z represented as the random separable of crayfish, this is expressed in Eq. (21),

$$z = round(rand \times (N - 1)) + 1 \quad (21)$$

At the competition phase, crayfish engage through one other, crayfish X_i modifies location depending on their position X_z of other crayfish. The location may be changed to increase the COA search range and improve the exploration capability of algorithm.

3.5.1.7. Step 7: foraging stage (Exploitation). The temperature is appropriate to the crayfish for eating when $temp \leq 30$. At the moment, the crayfish start to approach the food. Crayfish determine the food dimension after locating it. The location of food X_{food} is expressed in Eq. (22)

$$X_{food} = X_G \quad (22)$$

The crayfish therefore just has to approach the meal and consume it immediately, which is expressed in Eq. (23),

$$X_{i,j}^{t+1} = (X_{i,j}^t - X_{food}) \times p + p \times rand \times X_{i,j}^t \quad (23)$$

In the Foraging stage, crayfish adopt various feeding techniques depend on food size Q , food X_{food} signifies the ideal solution.

3.5.1.8. Step 8: termination. By this, the ideal parameter values of α , α and η regularization tenure constraint from SACGAN classifier are enhanced in terms of Crayfish Optimization Algorithm repeats until fulfill halting criteria $t = t + 1$. In the end, SACGAN—COA efficiently classifies the cloud data as common and invaders (DoS, Probe, R2L and U2R) with increased accuracy by decreasing the computation period with error.

4. Results with discussion

The experimental outcomes about proposed CybS-CC-SACGAN—COA are discussed to enhance security on the cloud environment. The experiments are performed in laptop along Intel core i7 CPU, 4GB RAM, 64-bit Windows 10 OS. NSL-KDD dataset is taken for carried out these experiments. The proposed technique is simulated in Python using several performance measures. The outcomes of proposed CybS-CC-SACGAN—COA are analyzed with existing methods like CybS-CC-BPNN-ArBCOA (Wen, 2022), CybS-CC-DKNN (Mayuranathan et al., 2022), CybS-CC-DNN-GTOA (Balamurugan et al., 2022) and CybS-CC-WDBiLSTM-GWO (Samunnisa et al., 2023) methods.

4.1. Dataset description

The efficiency of CybS-CC-SACGAN—COA is validated utilizing the NSL-KDD dataset. It includes 3,79,679 archives depicts four assaults including DoS, Probe, R2L, and U2R attacks, in addition to archives showing normal behavior (no attacks). In this, On behalf of training 50 % data are taken then the remaining 50 % data for testing.

4.2. Performance metrics

This is a main task to select the ideal classifier. To analyze the performance, the given confusion matrix is necessary.

- True Positive (TP): intrusive properly characterized as intrusive.
- True Negative (TN): normal properly characterized as normal.
- False Positive (FP): normal wrongly characterized as intrusive.
- False Negative (FN): intrusive wrongly characterized as normal.

4.2.1. Accuracy

The ratio of all correctly classified instances to all examples that exist is known as accuracy. It displays whether the proposed IDS has the capacity for raising appropriate alarms when identifying the attacks and not to create false alarms when normal the connection network instance. This is computed by Eq. (24),

$$Accuracy = \frac{(TP + TN)}{(TP + FP + TN + FN)} \quad (24)$$

4.2.2. Precision (P)

It shows the rate of positive predicted values that actually occurring is precision. P has a direct impact on the proposed IDS performance because a higher P value corresponds to a lower false positive rate and vice versa. It is expressed in Eq. (25),

$$Precision = \frac{TP}{(TP + FP)} \quad (25)$$

4.2.3. Recall

This is the ratio of all attacks that the IDS detected to all attacks that really occurred. It displays the capacity of proposed IDS for identifying the attacks. It is expressed in Eq. (26)

$$Recall = \frac{TP}{(TP + FN)} \quad (26)$$

4.2.4. F1 score

It evaluates the accurateness of the proposed IDS with regarding to precision and recall rate values. It is expressed in Eq. (27)

$$F1\ Score = \frac{TP}{\left(TP + \frac{1}{2} [FP + FN] \right)} \quad (27)$$

4.2.5. RoC

The RoC measures the performance of a categorization technique by quantifying its capability to distinguish among positive and negative classes across various threshold values. RoC assessed under eqn (28),

$$RoC = 0.5 \times \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (28)$$

4.2.6. False positive rate (FPR)

The ratio of all misclassified normal occurrences to all normal instances is displayed. FPR value shows how many false positive alarms the proposed IDS would produce. The FPR value must be as low as feasible to prevent the system administrator from becoming confused by too many false positive alarms. It is expressed in Eq. (29)

$$\text{False Positive Rate} = \frac{FP}{FP + TN} \quad (29)$$

4.2.7. False negative rate (FNR)

It can provide the ratio of total misclassified attack instances to the total attack instances. FNR value indicates how many false negative alarms the proposed IDS it would generate. To prevent too many attacks from going undetected, the FNR value must be as low as feasible. It is given in Eq. (30)

$$\text{False Negative Rate} = \frac{FN}{FN + TP} \quad (30)$$

4.2.8. Detection error rate (DER)

Detection Error Rate (DER) is a metric used to evaluate the performance of IDS. DER provides a comprehensive measure by considering both false positives and false negatives. It is defined as the sum of false positive rate as well as false negative rate. It is exhibited in Eq. (31),

$$\text{Detection Error Rate}(\%) = 100 - \text{Accuracy}(\%) \quad (31)$$

4.2.9. Detection latency

Detection latency is a measure of the time it takes for IDS to detect and respond to a security incident. It represents the elapsed time from the occurrence of a security event to the moment the IDS identifies and generates an alert or takes an action. It is given in Eq. (32)

$$\text{Detection Latency} = T_D - T_O \quad (32)$$

where, T_D is the time at which the IDS detects the security incident, T_O is the time at which the security event actually occurs.

4.2.10. Throughput

It measures the rate at which a system can complete a task or deliver a service over a specific period of time. In the context of network or communication systems, throughput often refers to the amount of data transferred successfully between two points in the network within a given timeframe. The general equation for throughput is given in Eq. (33)

$$\text{Throughput} = \frac{\text{Amount of Data transferred}}{\text{Time taken}} \quad (33)$$

4.2.11. Scalability

Scalability represents the ability of the system to handle an increasing volume of network traffic and data while maintaining its effectiveness. It assesses how well the IDS can expand its capacity to accommodate growth without a significant degradation in performance. It is given in Eq. (34)

$$\text{Scalability} = \frac{LW}{CW} \times 100 \quad (34)$$

where, LW is the Performance Measure for Larger Workload, CW is the Performance Measure for Current Workload.

4.2.12. Resource consumption

Resource consumption in the context of intrusion detection refers to

the amount of computing resources, such as CPU, memory, and network bandwidth, utilized by the intrusion detection system (IDS) to perform its tasks. Managing resource consumption is crucial to ensure that the IDS effectively detects and responds to security threats without negatively impacting the overall performance of the cloud or network environment. It is expressed in Eq. (35)

$$\begin{aligned} \text{Response Consumption} = & \text{CPU}_{\text{usage}} + \text{Memory}_{\text{usage}} + \text{Network Bandwidth}_{\text{usage}} \\ & + \text{Other resource}_{\text{usage}} \end{aligned} \quad (35)$$

4.2.13. Computation time

Computation time, also known as execution time or processing time, it measures the amount of time it takes for a specific task, algorithm, or process to complete its execution. It is a crucial metric for assessing the efficiency and speed of a computational system. The computation time is typically measured in seconds. It is given in Eq. (36)

$$\text{Computation Time} = \text{End Time} - \text{Start Time} \quad (36)$$

4.3. Performance analysis

Table 4-7 and Fig. 3-7 represents the simulation outcomes of proposed CybS-CC-SACGAN—COA approach. The proposed CybS-CC-SACGAN—COA approach is related with existing CybS-CC-BPNN-ArBCOA (Wen, 2022), CybS-CC-DKNN (Mayuranathan et al., 2022), CybS-CC-DNN-GTOA (Balamurugan et al., 2022) and CybS-CC-WDBiLSTM-GWO (Geetha and Deepa, 2022) systems.

Table 4 tabulates Accuracy and Precision analysis. It measures the rate of exact detected instances to the total instances at the database. It is expressed as a percentage. Here, the proposed CybS-CC-SACGAN—COA technique achieves maximum accuracy value over the existing techniques. The CybS-CC-BPNN-ArBCOA technique achieves lesser accuracy value, the CybS-CC-DNN-GTOA method achieves slightly better results with maximized accuracy value, the CybS-CC-WDBiLSTM-GWO method attains moderate accuracy value, the CybS-CC-DKNN method achieves near ideal identification accuracy value. But, the proposed CybS-CC-SACGAN—COA attains 22.41 %, 18.75 %, 26.06 %, and 21.20 % higher

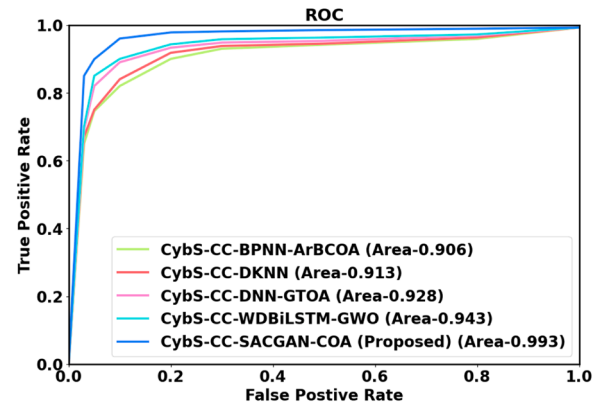


Fig. 3. RoC Analysis.

Table 4
Accuracy and Precision analysis.

Models	Accuracy (%)					Precision (%)				
	DoS	Probe	R2L	U2R	Normal	DoS	Probe	R2L	U2R	Normal
CybS-CC-BPNN-ArBCOA	80.2	83.5	86.4	86.1	83.9	73.4	80.3	76.3	87.6	79.2
CybS-CC-DKNN	85	75.8	70.9	89	86.2	82.6	81	77.5	88.2	77.6
CybS-CC-DNN-GTOA	86.3	88.5	77.2	80.4	81.7	70.3	70	84.2	84.2	83.5
CybS-CC-WDBiLSTM-GWO	87.2	84.2	78.7	71.9	87.8	81.1	84	72.3	81.2	86.1
CybS-CC-SACGAN—COA (Proposed)	99.35	99.5	99.56	99.44	99.76	98.8	98.6	99.3	98.6	98.8

accuracy for DoS Attack; 27.24 %, 29.47 %, 15.96 % and 22.04 % higher accuracy for Probe Attack; 19.48 %, 24.01 %, 21.38 % and 18.42 % higher accuracy for R2L Attack; 24.01 %, 28.47 %, 16.42 % and 26.67 % higher accuracy for U2R Attack; 22.60 %, 34.73 %, 18.44 % and 30.25 % higher accuracy for Normal over existing CybS-CC-BPNN-ArBCOA, CybS-CC-DKNN, CybS-CC-DNN-GTOA and CybS-CC-WDBiLSTM-GWO respectively.

Precision is fraction of real anomalous activities detected as Usual among identified activities detected as Usual. It is expressed as a percentage. The proposed CybS-CC-SACGAN—COA technique achieves maximum precision value over the existing techniques. The CybS-CC-WDBiLSTM-GWO technique achieves lesser precision value, the CybS-CC-DKNN method achieves slightly better results with maximum precision value, the CybS-CC-BPNN-ArBCOA technique achieves moderate precision value, the CybS-CC-DNN-GTOA method achieves near ideal precision value. The proposed CybS-CC-SACGAN—COA method attains 19.08 %, 29.63 %, 31.05 %, and 35.45 % greater precision for DoS Attack; 25.10 %, 15.80 %, 28.43 %, 21.30 % higher precision for Probe Attack; 27.04 %, 21.42 %, 19.45 % and 32.71 % greater precision for R2L Attack; 26.61 %, 24.47 %, 19.12 % and 30.27 % greater precision for U2R Attack; 20.23 %, 28.42 %, 19.74 % and 27.63 % greater precision for Normal analyzed to the existing CybS-CC-BPNN-ArBCOA, CybS-CC-DKNN, CybS-CC-DNN-GTOA and CybS-CC-WDBiLSTM-GWO models.

Table 5 shows Performance analysis of Recall and F1-Score. Recall is particularly important in situations where the positive instance cost of missing is higher. For eg, in the context of intrusion detection, a false negative would indicate that a genuine security threat was undetected, which could have detrimental effects. Therefore, a higher recall specifies that the system is better at capturing most of the positive instances, minimizing the number of undetected instances. Here, the proposed CybS-CC-SACGAN—COA method attains maximum Recall value over existing techniques. The CybS-CC-WDBiLSTM-GWO technique achieves lesser Recall value, the CybS-CC-DKNN method achieves slightly better results with maximized Recall value, the CybS-CC-DNN-GTOA technique attains moderate Recall value, the CybS-CC-BPNN-ArBCOA method achieves near ideal Recall value. But, the proposed CybS-CC-SACGAN—COA attains 24.16 %, 27.14 %, 18.42 %, and 19.03 % greater Recall for DoS Attack; 22.48 %, 17.15 %, 20.19 % and 18.42 % higher Recall for Probe Attack; 23.62 %, 18.48 %, 25.78 % and 22.97 % higher Recall for R2L Attack; 21.26 %, 24.18 %, 16.46 % and 32.06 % higher Recall for U2R Attack; 25.15 %, 24.13 %, 20.84 % and 19.08 % greater Recall for Normal compared with existing methods like CybS-CC-BPNN-ArBCOA, CybS-CC-DKNN, CybS-CC-DNN-GTOA and CybS-CC-WDBiLSTM-GWO respectively.

F1-Score is a harmonic mean of precision or recall. Here, the CybS-CC-SACGAN—COA achieves maximum F1-Score value over the existing techniques. The CybS-CC-BPNN-ArBCOA technique achieves lesser F1-score value, the CybS-CC-DNN-GTOA method achieves slightly better outcome with maximized F1-score value, the CybS-CC-DKNN technique attains moderate F1-score value, the CybS-CC-WDBiLSTM-GWO method achieves near ideal F1-score value. But, the proposed CybS-CC-SACGAN—COA attains 22.47 %, 16.40 %, 23.90 %, and 19.15 % higher F1-score for DoS Attack; 18.32 %, 28.84 %, 20.12 % and 22.35 % greater F1-score for Probe Attack; 25.64 %, 19.15 %, 26.90 % and 21.05 %

Table 6

Performance analysis of Detection Error Rate.

Methods	Detection Error Rate (%)				
	DoS	Probe	R2L	U2R	Normal
CybS-CC-BPNN-ArBCOA	19.8	16.5	13.6	13.9	16.1
CybS-CC-DKNN	15	24.2	29.1	11	13.8
CybS-CC-DNN-GTOA	13.7	11.5	22.8	19.6	18.3
CybS-CC-WDBiLSTM-GWO	12.8	15.8	21.3	28.1	12.2
CybS-CC-SACGAN—COA (Proposed)	0.65	0.5	0.44	0.56	1

greater F1-score for R2L Attack; 27.20 %, 20.48 %, 25.71 % and 31.61 % greater F1-score for U2R Attack; 19.90 %, 24.54 %, 20.32 % and 28.72 % greater F1-score for Normal compared with existing CybS-CC-BPNN-ArBCOA, CybS-CC-DKNN, CybS-CC-DNN-GTOA and CybS-CC-WDBiLSTM-GWO models.

Table 6 shows the Performance analysis of Detection Error Rate. DER takes the average of false acceptance rate along false rejection rate, essentially calculating the midpoint between the two error rates. A lower DER value indicates better overall performance of the biometric system, as it implies a better balance between false acceptances and false rejections. It is expressed in percentage. Here, the CybS-CC-SACGAN—COA method attains least detection error rate value over the existing techniques. The proposed CybS-CC-SACGAN—COA technique achieves 27.05 %, 19.30 %, 24.70 %, and 21.52 % lower detection error rate for DoS Attack; 31.27 %, 25.51 %, 20.32 % and 16.78 % lower detection error rate for Probe Attack; 24.81 %, 26.42 %, 18.14 % and 27.78 % detection lower error rate for R2L Attack; 25.30 %, 24.49 %, 14.09 % and 28.20 % lesser detection error rate for U2R Attack; 31.12 %, 19.98 %, 25.94 % and 92.41 % lesser detection error rate for Normal compared with existing method like CybS-CC-BPNN-ArBCOA, CybS-CC-DKNN, CybS-CC-DNN-GTOA and CybS-CC-WDBiLSTM-GWO models.

Table 7 displays Computation Time and Detection Latency. Computation time is a key factor in assessing the efficiency of algorithms and programs. A faster computation time often indicates a more efficient and responsive system. Here, the CybS-CC-SACGAN—COA technique achieves least computation time value over the existing techniques. The proposed CybS-CC-SACGAN—COA attains 22.52 %, 27.30 %, 32.18 % and 20.22 % lower computation time compared with existing CybS-CC-BPNN-ArBCOA, CybS-CC-DKNN, CybS-CC-DNN-GTOA and CybS-CC-WDBiLSTM-GWO models.

Detection latency is critical for timely response to security incidents. Rapid detection allows security teams to respond quickly, potentially mitigating the impact of an attack or preventing further damage. Here,

Table 7

Performance analysis of Computation Time and Detection Latency.

Methods	Computation Time (s)	Detection Latency (s)
CybS-CC-BPNN-ArBCOA	265	17.5
CybS-CC-DKNN	278	21.3
CybS-CC-DNN-GTOA	207	19.5
CybS-CC-WDBiLSTM-GWO	218	15.25
CybS-CC-SACGAN—COA (Proposed)	95	1.25

Table 5

Performance analysis of Recall and F1-Score.

Methods	Recall (%)					F1-Score (%)				
	DoS	Probe	R2L	U2R	Normal	DoS	Probe	R2L	U2R	Normal
CybS-CC-BPNN-ArBCOA	78.4	71.5	81.8	81.2	86.1	87	87.2	85.3	76.6	88.6
CybS-CC-DKNN	87.3	76.3	84.7	82.8	86.9	89.8	71.5	70.3	72.1	70.6
CybS-CC-DNN-GTOA	77.9	74.1	79.6	75.7	89.1	81.7	88.1	79.5	73.6	81.6
CybS-CC-WDBiLSTM-GWO	84.4	81.6	70.8	88.8	85	74.7	75.6	81.4	70.4	80
CybS-CC-SACGAN—COA (Proposed)	98.6	99.3	98.5	99.4	98.7	99.3	98.5	99.2	99.1	99.5

the CyBS-CC-SACGAN—COA technique achieves least Detection latency value over the existing techniques. The proposed CyBS-CC-SACGAN—COA attains 29.52 %, 27.30 %, 23.18 % and 10.22 % lower Detection latency compared with existing Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO models.

AUC-ROC measures the entire presentation of a SACGAN by computing the area under the ROC curve. The proposed Cybs-CC-SACGAN—CO achieves maximal RoC value over the existing techniques. The Cybs-CC-BPNN-ArBCOA techniques achieves least RoC value, the Cybs-CC-DKNN method achieves slightly better outcome with maximal RoC value, the Cybs-CC-DNN-GTOA attains minimal RoC value, the Cybs-CC-WDBiLSTM-GWO method achieves near ideal value. But, the proposed Cybs-CC-SACGAN—CO achieves 18.24 %, 19.47 %, 22.07 % and 17.02 % greater RoC for Normal compared with existing Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO models.

Resource consumption is a vital performance metric for IDSs, ensuring their practical viability and sustainability in real-world environments. Effective IDS design must balance detection accuracy with resource efficiency to maintain system performance, scalability, and cost-effectiveness. Fig. 4 shows Resource Consumption. Here, the proposed Cybs-CC-SACGAN—COA technique attains 12.98 %, 23.86 %, 29 % and 30.1 % lower Resource consumption analyzed to the existing Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO models respectively.

Scalability is crucial for ensuring that a system can accommodate growth without a significant degradation in performance. Several scalability metrics are commonly used to evaluate different aspects of a system's ability to scale. Fig. 5 shows the performance analysis of Scalability. Here, the Cybs-CC-SACGAN—COA technique attains 22.98 %, 20.86 %, 29.34 % and 10.1 % higher Scalability evaluated to the existing Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO models respectively.

A count of data that can be transferred successfully over a network in a specific time frame is throughput. Fig. 6 shows the throughput analysis. Here, the proposed Cybs-CC-SACGAN—COA technique attains 28.98 %, 21.86 %, 24.34 % and 19.1 % higher throughput analyzed to the existing Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO models respectively.

FPR provides insights into the SACGAN's tendency to generate false alarms, specifically for the negative class. A lower FPR indicates that the model is better at avoiding false positive errors, which is generally desirable in applications where false positives have higher consequences. Fig. 7 shows the Performance analysis of False Positive Rate. Here, the Cybs-CC-SACGAN—COA method attains least FPR value over the existing techniques. The proposed Cybs-CC-SACGAN—COA technique achieves 27.05 %, 29.30 %, 24.70 %, and 21.52 % lower FPR for DoS Attack; 31.27 %, 25.51 %, 20.32 % and 26.78 % lower FPR for

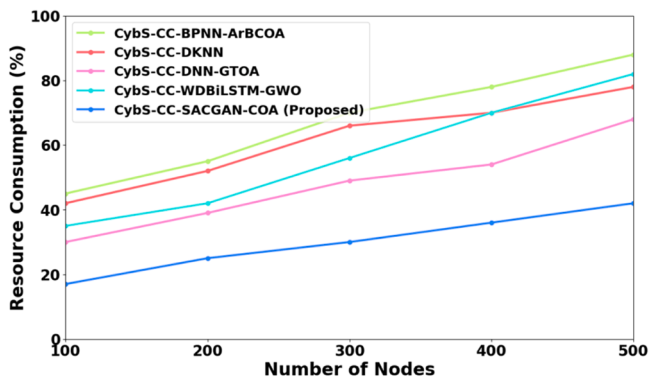


Fig. 4. Resource Consumption Analysis.

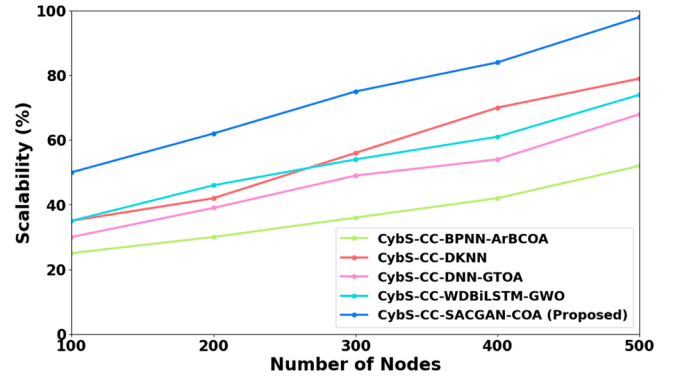


Fig. 5. Performance analysis of Scalability.

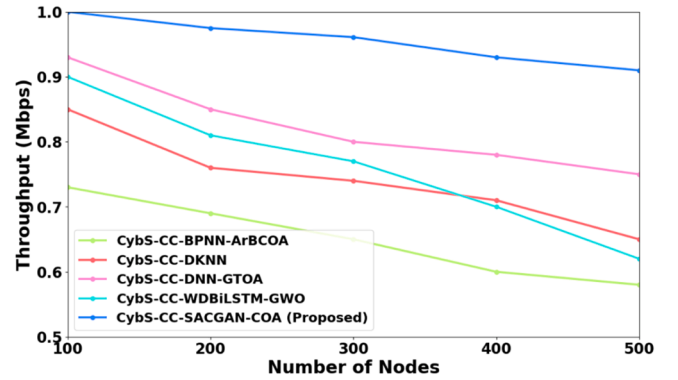


Fig. 6. Performance analysis of Throughput.

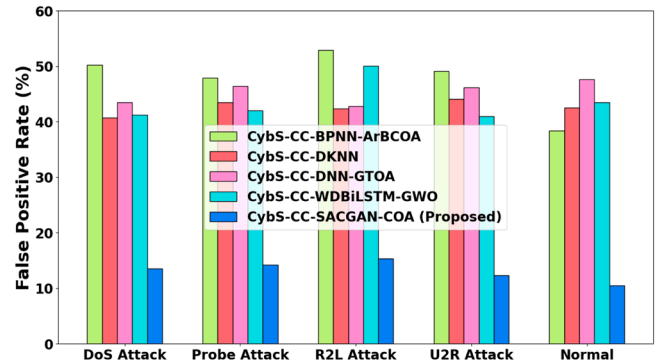


Fig. 7. Performance analysis of False Positive Rate.

Probe Attack; 24.81 %, 26.42 %, 28.14 % and 27.78 % lower FPR for R2L Attack; 25.30 %, 24.49 %, 24.09 % and 28.20 % lesser FPR for U2R Attack; 21.12 %, 29.98 %, 25.94 % and 22.41 % lesser FPR for Normal compared with existing method like Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO models.

4.4. Discussion

This paper proposes SACGAN optimised with Crayfish Optimization Algorithm to optimize SACGAN network algorithm. SACGAN model classifies the input data into DoS, Probe, R2L, U2R attacks and normal. The proposed method is contrasted with four other existing methodologies; they are Cybs-CC-BPNN-ArBCOA, Cybs-CC-DKNN, Cybs-CC-DNN-GTOA and Cybs-CC-WDBiLSTM-GWO. These models were tested on 37,129 samples of the NSL-KDD dataset after being trained on 1,11,386 instances. Performance of intrusion detection accuracy is the

primary evaluation. The experimental outcomes depicts that the identification of cloud computing intrusion attack samples, the proposed CyBS-CC-SACGAN—COA approach has a high detection rate of 99.67 % and can successfully separate aberrant data from normal data. Furthermore, the average false alarm rate is a low 3.42 %. This proposed method has a higher RoC value of 0.9915, which is 2.8 % to 5.8 % higher than other models. Finally, the outcomes depict that the proposed technique identifying anomaly data effectually in Cloud Computing to improve Cyber Security.

5. Conclusion

In this paper, CyBS-CC-SACGAN—COA is proposed. The relevant features were selected for improving the training speed and decreasing the computational time of the technique. The proposed CyBS-CC-SACGAN—COA approach is executed in Python under performance metrics. The proposed CyBS-CC-SACGAN—COA approach attains 19.08 %, 29.63 %, 31.05 %, and 35.45 % higher Precision; 24.16 %, 27.14 %, 18.42 %, and 19.03 % higher recall; 21.15 %, 25.29 %, 26.15 %, and 30.46 % higher throughput than the existing methods, like CyBS-CC-BPNN-ArBCOA, CyBS-CC-DKNN, CyBS-CC-DNN-GTOA and CyBS-CC-WDBiLSTM-GWO respectively.

While the proposed anomaly-based NIDS utilizing Crayfish Optimization Algorithm and Self-Attention Conditional Generative Adversarial Network has demonstrated promising results, it is significant to acknowledge some limitations inherent in the current study that is, the efficacy of the proposed NIDS is contingent on the quality and representativeness of the training dataset (NSL-KDD). The inherent biases and limitations within the dataset may impact the generalizability of the model to real-world cloud environments. The proposed IDS assume a static network environment during the training and testing phases. The impact of dynamic network conditions, such as fluctuations in traffic volume and topology changes, on the system's performance needs further investigation.

Conducting experiments in real-world cloud environments will provide valuable insights into the adaptability and performance of the proposed NIDS in practical scenarios. This involves collaboration with cloud service providers for data access and validation. Exploring the integration of multiple optimization techniques, beyond COA, to optimize the control parameters of SACGAN. This may involve experimenting with hybrid optimizations, transfer learning techniques etc.

CRedit authorship contribution statement

Sahaya Stalin Jose. G: Methodology. **G. Sugitha:** Supervision. **Ayshwarya Lakshmi. S:** Supervision. **Preethi Bangalore Chaluvaraj:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

No data was used for the research described in the article.

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